Nawa Dahal

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Education

San José State University — San José, California

B.S. Applied and Computational Mathematics — Aug. 2025 – Expected May 2027

GPA: 3.93/4.0

De Anza College — Cupertino, California

Associate of Science - AS, Mathematics — Sep. 2022 – Jun. 2025

GPA: 3.78/4.00

Research Interests

Numerical Analysis, Scientific Computing, Nonlinear Partial Differential Equations, Applied Analysis, Dynamical Systems, and Mathematical Modeling.

Research Experience

A Matrix-Based Perspective on Multi-Level Richardson Extrapolation — San José State University

Undergraduate Researcher | Advisor: Dr. Mohammad Saleem — Sep. 2025 – Dec. 2025

- Formulated multi-level Richardson extrapolation through a Vandermonde-type linear system for the extrapolation weights.
- Identified the weights as Lagrange interpolation basis polynomials evaluated at zero, connecting extrapolation with polynomial interpolation.
- Analyzed conditioning, extrapolation-weight growth, and floating-point error using MATLAB experiments with smooth, insufficiently smooth, and highly oscillatory integrands.

Analysis and Numerical Investigation of Nonlinear Semiconductor Drift–Diffusion Systems — San José State University

Honors Undergraduate Researcher | Advisor: Dr. Daniel Brinkman — Jan. 2026 – Jul. 2026

- Derived second-order finite-difference discretizations of the equilibrium Poisson–Boltzmann equation and steady-state drift–diffusion–Poisson system.
- Analyzed fixed-point and quasi-linearized iterations using contraction arguments and spectral properties of the discrete Laplacian; proved that the quasi-linearized update is equivalent to Newton's method.
- Developed a depletion-aware initial approximation and implemented Gummel iteration with Scharfetter–Gummel fluxes, Shockley–Read–Hall recombination, and Ohmic contact conditions.
- Studied convergence and current conservation under varying doping and applied bias, and computed current–voltage characteristics consistent with the Shockley diode model.

Photovoltaic Extension of a Drift–Diffusion–Poisson Solver — San José, California

Independent Researcher — Summer 2026

- Extended the one-dimensional drift–diffusion–Poisson solver from dark diode transport to illuminated solar-cell simulation by incorporating Beer–Lambert optical generation into the carrier-continuity equations.
- Computed illuminated current–voltage characteristics through terminal-voltage sweeps and extracted the short-circuit current density, open-circuit voltage, maximum-power point, and fill factor.

Two-Dimensional Semiconductor and PNP Transistor Modeling — San José State University

Undergraduate Researcher | Advisor: Dr. Daniel Brinkman — Aug. 2026 – Present

- Extended the steady-state drift–diffusion–Poisson framework from one to two spatial dimensions using finite differences, sparse linear systems, and mixed Ohmic-contact and insulating boundary conditions.
- Developed a planar PNP transistor simulation with separate emitter, base, and collector contacts, using physics-informed initialization and applied-voltage continuation to obtain convergence and examine terminal-current balance.
- Next stage: investigate a differential-algebraic formulation of the coupled system as an alternative to the linearized Poisson subproblem within Gummel iteration.

Teaching Experience

MATH 19: Precalculus — San José State University

Instructional Student Assistant: Workshop Facilitator — Aug. 2025 – Dec. 2025

- Independently led weekly workshop sessions for approximately 25 students, guiding collaborative problem solving and mathematical discussion.
- Administered and graded student assessments and provided feedback on students' mathematical reasoning.

MATH 33LA: Differential Equations and Linear Algebra — San José State University

Instructional Student Assistant: Grader — Jan. 2026 – May 2026

- Graded quizzes, midterms, and the final examination for a large-enrollment course of approximately 160 students.
- Evaluated students' solutions using detailed rubrics, identified conceptual and computational errors, and provided written feedback.

MATH 71W: Calculus Workshop for Business and Aviation — San José State University

Instructional Student Assistant: Workshop Facilitator — Aug. 2026 – Present

- Lead weekly workshop sessions and guide students through collaborative problem solving in applied calculus.
- Administer and grade student assessments and reinforce students' understanding through individual and group instruction.

MATH 143C: Numerical Analysis and Scientific Computing — San José State University

Instructional Student Assistant: Grader — Aug. 2026 – Present

- Grade homework and computational projects for an upper-division numerical analysis course.
- Evaluate mathematical derivations, numerical algorithms, and computational results for accuracy and clarity.

Publications

Dahal, N. (2026). A matrix-based perspective on multi-level Richardson extrapolation. *SIAM Undergraduate Research Online, 19*, 409–428. https://doi.org/10.1137/26S1857207

Honors Thesis

Analysis and Numerical Investigation of Nonlinear Semiconductor Drift–Diffusion Systems — 2026

Undergraduate honors thesis, Department of Mathematics and Statistics, San José State University. Advisor: Dr. Daniel Brinkman.

Posters & Presentations

Numerical Investigation of the Poisson–Boltzmann Equation

Research Poster

- Presented at the **College of Science Student Research Day** (Apr. 17, 2026).
- Presented at the **3rd Annual SIAM Northern and Central California Section Conference (NCC26)**, UC Davis (Oct. 27, 2026).

Honors & Awards

2026 **SIAM NCC26 Student Travel Award,** SIAM NCC26 Organizing Committee — UC Davis

2022–2025 **Dean's List (Every Quarter Enrolled),** De Anza College — Cupertino, California

Fall 2025 **President's Scholar,** San José State University — San José, California

Relevant Coursework

In Progress — Partial Differential Equations, Complex Analysis, Advanced Dynamical Systems

Completed — Introduction to Analysis, Advanced Calculus, Ordinary Differential Equations and Dynamical Systems, Numerical Analysis I, Numerical Analysis II, Applied Probability and Statistics

Planned Coursework Applied Mathematics I (Graduate Partial Differential Equations), Numerical Solution of PDEs

Technical Skills

MATLAB | C++ | LaTeX